

数据结构与算法

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Chapter 1: Arrays and Linked Lists

Arrays store elements in contiguous memory locations, enabling $O(1)$ random access.

Linked Lists connect nodes via pointers, enabling $O(1)$ insertion and deletion.

Complexity comparison:

- Array access: $O(1)$
- Array insert: $O(n)$
- List access: $O(n)$
- List insert: $O(1)$

Chapter 2: Stacks and Queues

Stack (LIFO - Last In, First Out):

- push(x) - add to top
- pop() - remove from top
- peek() - view top element

Queue (FIFO - First In, First Out):

- enqueue(x) - add to back
- dequeue() - remove from front

Applications:

- Stack: function calls, expression evaluation
- Queue: BFS, task scheduling

Chapter 3: Sorting Algorithms

Quick Sort: Average $O(n \log n)$, divide and conquer strategy.

Merge Sort: $O(n \log n)$ guaranteed, stable sort.

Bubble Sort: $O(n^2)$, simple but inefficient.

Heap Sort: $O(n \log n)$, in-place sorting.

For most practical cases, Quick Sort is preferred due to its cache efficiency.